



7 The curve C has polar equation $r^2 = e^{\sin \theta} \cos \theta$, for $-\frac{1}{2}\pi \leq \theta \leq \frac{1}{2}\pi$.

- (a) Find the polar coordinates of the point on C that is furthest from the pole, giving your answers correct to 3 decimal places. [5]

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- (b) Find the polar coordinates of the point on C that is furthest from the half-line $\theta = \frac{1}{2}\pi$, giving your answers correct to 3 decimal places. [5]

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(c) Sketch C .

[3]

(d) Find the area of the region bounded by C , giving your answer in exact form.

[3]

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